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Diastereoselective Construction of Cyclopent-2-enone-4-ols from Aldehydes and 1,2-Allenones Catalyzed by *N*-Heterocyclic Carbene

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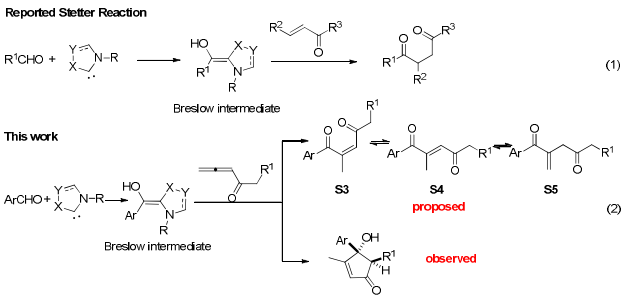
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Dengke Ma, Chunling Fu and Shengming Ma*

Cyclopent-2-enone-4-ols have been prepared highly diastereoselectively from readily available aromatic aldehydes and 1,2-allenones catalyzed by *N*-heterocyclic carbene. Mechanistic studies showed that there are four possible pathways.

Cyclopent-2-enone-4-ol unit exists in some natural products and biological active compounds,[1] thus, efficient synthetic routes to these compounds are of high interest.[2] Known methods for their preparations from acyclic precursors includes: cyclizations of 1,2-diketones with 3-oxocarboxylic acid esters or amides,[3] ketones,[4] 1,3-bis(trimethylsiloxy)-1,3-dienes,[3a,5] 2-(trimethylsiloxy)allyltriemthylsilane,[6] or 2,3-allenoates;[7] transition metal-catalyzed intramolecular coupling of ethyl 3-(triethylsilyloxy)-5-(tributylstannyl)pent-4(*Z*)-enthioates followed by desilylation;[8] ring closing metathesis of 5-OTBS or DPS-hepta-1,6-dien-3-ones followed by possible deprotection;[9] Tandem Aldol–Brook reaction of β-(phenylthio)acryloyl silanes with lithium enolates followed by simultaneous desilylation and dephenylthiolation;[10] tandem ozonolysis–Knoevenagel cyclization of *tert*-butyl 5-OTBS-3-oxohept-6-enoates;[11] Nazarov reaction and desiylation of (*E*)-4-methoxymethyl-1-silyloxyhexa-1,4,5-trien-3-ols;[12] and the cyclization reactions involving *cis*-2-en-1,4-diketones[13] or (*Z*)-4-oxopent-2-enal.[14]

On the other hand, traditionally, the treatment of aldehydes with *N*-heterocyclic carbenes would afford the Breslow intermediate, subsequent Michael addition with α,β-unsaturated carbonyl compounds gave 1,4-diketones (the Stetter reaction) (Scheme 1, eq 1).[15,16] We hypothesized that when 1,2-allenone is treated with the Breslow intermediate, a mixture of C-C double bonds regio-isomers 1,4-diketones **S3**, **S4** and **S5**, could be produced. Rather unexpectedly, the cyclic 4-hydroxycyclopentenone product was formed highly diastereoselectively (Scheme 1, eq 2).



Scheme 1 Stetter reaction and this reaction.

In the beginning, we attempted the reaction of 4-bromobenzaldehyde **1a** with 1,2-allenyl ketone **2a** using Cs_2CO_3 as the base and toluene as the solvent in the presence of different commercially available *N*-hetereocyclic carbene precursors **A** to **E**. In addition to the C=C bond regio-isomer **4aa**, (*E*)-**3aa**, and (*Z*)-**3aa**, a new cyclic product **5aa** (dr = 95:5) was observed in 40% yield when changing the solvent from toluene to dioxane (Table S1, entry 10, see ESI†).

Inspired by the observation above, reaction parameters including the loading of the NHC precursor **A1** and Cs_2CO_3, solvents, together with the reaction temperature were screened extensively. As a result, increasing the loading of Cs_2CO_3 led to a complete transformation to product **5aa**. However, the diastereoselectivity was not satisfactory (Table 1, entry 1). Running the reaction at 10 °C gave a better diastereoselectivity with a lower yield (Table 1, entry 3). By reducing the loading of base, a slightly higher diastereoselectivity was observed with a lower yield (Table 1, entries 4 and 5). As expected, the reaction at 50 °C afforded the product **5aa** with a lower diastereoselectivity (81:19), although the yield is 68% (Table 1, entry 6). After further optimization, the best diastereoselectivity was obtained with 10 mol % **A1** and 25 mol % Cs_2CO_3 using dioxane as solvent at 15 °C (Table 1, entry 8). Other solvents, such as DMF, THF and CH_3CN didn't give a better result (Table 1, entries 11-13).

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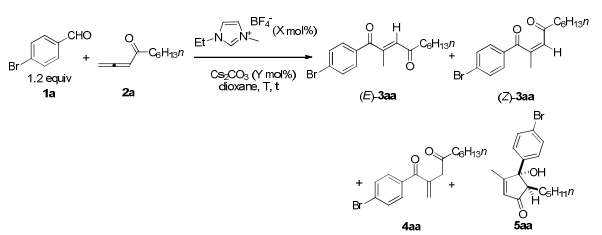
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Table 1 Optimization of the reaction conditions[a]

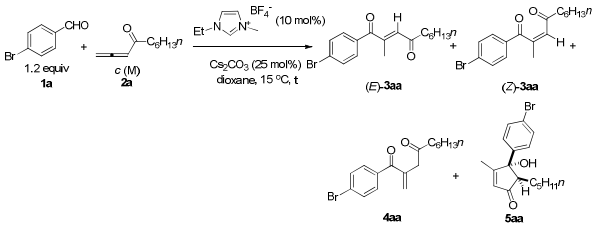


Entry	X	Y	T (°C)	t (h)	NMR Yield% (E)-3aa	(Z)-3aa	4aa	5aa (dr)[b]
1	10	30	21	17	2	—	—	61 (90:10)
2	10	30	15	11.75	—	—	—	57 (89:11)
3	10	30	10	21.75	4	<1	<1	49 (94:6)
4	10	20	20	22	2	1	<0.5	63 (91:9)
5	10	15	20	22	4	1	<1	45 (93:7)
6	10	20	50	22	—	—	—	68 (81:19)
7	10	20	15	23	8	1	1	55 (94:6)
8	10	25	15	23	11	1	2	51 (95:5)
9	10	25	15	47.5	3	<1	<0.5	49 (93:7)
10	12	25	15	22.3	4	—	<1	55 (93:7)
11[c]	10	20	20	21.75	—	—	—	—
12[d]	10	20	20	22	3	—	<0.5	38 (94:6)
13[e]	10	20	20	22	—	—	—	—

[a] Reaction conditions: A solution of NHC precursor and Cs_2CO_3 in dioxane (1 mL) was stirred at rt for 10 min, then **1a**, 0.4 mmol of **2a**, and dioxane (1 mL) were added sequentially. The resulting mixture was then stirred at T (°C) for the time indicated in the Table. [b] Determined by NMR of the crude product. [c] DMF was used as solvent. [d] THF was used as solvent. [e] CH_3CN was used as solvent.

To our delight, running the reaction at a concentration of 0.1 M for 1,2-allenone **2a** proved to be the best condition for both the yield and the diastereoselectivity (Table 2, entry 3).

Table 2 Effect of Concentration[a]

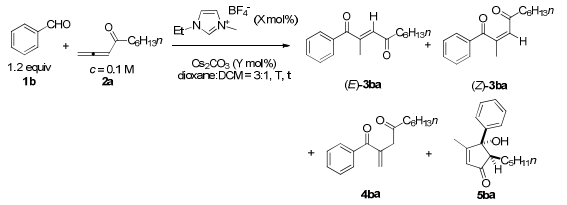


Entry	c (M)	t (h)	NMR Yield% (E)- 3aa	(Z)-3aa	4aa	5aa (dr)[b]
1	0.2	23	11	1	2	51 (95:5)
2	0.13	21.3	—	—	—	67 (88:12)
3	0.1	22.5	3	<1	<0.5	71 (94:6)
4	0.08	21.75	15	2	3	53 (94:6)

[a] Reaction conditions: A solution of NHC precursor and Cs_2CO_3 in dioxane (half) was stirred at rt for 10 min, then **1a**, 0.4 mmol of **2a**, and dioxane (half) were added sequentially. The resulting mixture was then stirred at 15 °C for the time indicated in the Table. [b] Determined by NMR of the crude product.

However, when benzaldehyde was applied under the optimal conditions illustrated as entry 3 in Table 2 even at a lower temperature of 13 °C, disappointingly, the diastereoselectivity dropped to 80:20 (Table 3, entry 1). In order to run the reaction at a lower temperature for better dr, we considered to add CH_2Cl_2 as the co-solvent: reaction at -10 °C under a catalyst loading of 20 mol % for **A1** and Cs_2CO_3 respectively in a mixed solvent of dioxane and CH_2Cl_2 (3/1) afforded 72% of the product and dr value was as high as 97:3 (Table 3, entry 6). Thus, this has been defined as the optimized conditions.

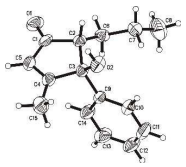
Table 3 Further optimization using benzaldehyde[a]



Entry	X	Y	T (°C)	t (h)	NMR Yield% (E)-3ba	(Z)-3ba	4ba	5ba (dr)[b]
1[c,d]	10	25	13	20.5	—	—	—	83 (80:20)
2	10	25	0	22	48	11	11	11 (100:0)
3	15	25	0	22.3	36	9	9	27 (100:0)
4[c]	20	25	0	12.25	—	—	—	82 (88:12)
5	20	25	-10	8.5	2	—	—	76 (96:4)
6	20	20	-10	8.5	7	1	1	72 (97:3)

[a] Reaction conditions: A solution of NHC precursor and Cs_2CO_3 in mixed solvent (2 mL) was stirred at rt for 10 min, and then **1b**, 0.4 mmol of **2a**, and solvent (2 mL) were added sequentially. The resulting mixture was then stirred at T for the time indicated in the Table. [b] Determined by NMR of the crude product. [c] The reaction was conducted on 1 mmol of **2a**. [d] Only dioxane was used as the solvent.

Gladly, aromatic aldehydes substituted with electron-withdrawing *p*-Br (Table 4, entry 1) and *p*-Ph (Table 4, entry 6) or electron-donating *p*-CH_3 (Table 4, entry 7) were well tolerated under the standard conditions. In addition, 2-naphthaldehyde was also compatible (Table 4, entries 8 and 9). When R^2 group was hydrogen atom, R^1 group could be linear alkyl (Table 4, entries 1-8) or branched alkyl (Table 4, entry 9). What's more, the reaction of allenone containing removable OBn group proceeded smoothly, which provided an opportunity for further elaboration (Table 4, entry 5). When R^1 was benzyl group, the dr values are slightly lower (Table 4, entries 4 and 8). Spirobicyclic compound **5bf** could be formed in 50% yield when propadienyl cyclohexylketone was used (Table 4, entry 10). The structure of the product was further confirmed by X-ray single crystal diffraction study of 4*R**,5*R**-**5bb**[17] (Figure 1).



4*R**,5*R**-**5bb**

Figure 1 ORTEP representation of 4*R**,5*R**-**5bb**.

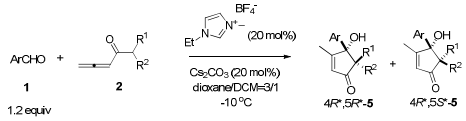
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Table 4 Substrates scope[a]



Entry	Ar/R^1/R^2	Yield of **5**%	Dr (4*R**,5*R**-**5**: 4*R**,5*S**-**5**)[b]
		NMR/isolated	Before/After separation
1	4-BrC_6H_4 (**1a**)/*n*-C_5H_{11}/H (**2a**)	54/53 (**5aa**)	95:5/>99:1
2	Ph (**1b**)/*n*-C_5H_{11}/H (**2a**)	80/76 (**5ba**)	96:4/99:1
3	Ph (**1b**)/*n*-C_3H_7/H (**2b**)	75/63 (**5bb**)	95:5/99:1
4	Ph (**1b**)/Bn/H (**2c**)	62/57 (**5bc**)	92:8/92:8
5	Ph (**1b**)/-$(CH_2)_4$OBn/H (**2d**)	74/65 (**5bd**)	96:4/99:1
6	4-PhC_6H_4 (**1c**)/*n*-C_5H_{11}/H (**2a**)	78/75 (**5ca**)[c]	94:6/96:4
7	4-MeC_6H_4 (**1d**)/*n*-C_3H_7/H (**2b**)	76/65 (**5db**)	94:6/99:1
8	2-naphthyl (**1e**)/Bn/H (**2c**)	60/57 (**5ec**)	92:8/91:9
9	2-naphthyl (**1e**)/*i*-propyl/H (**2e**)	64/56 (**5ee**)[d]	95:5/98:2
10[e]	Ph (**1b**)/-$(CH_2)_5$- (**2f**)	50/50 (**5bf**)	—

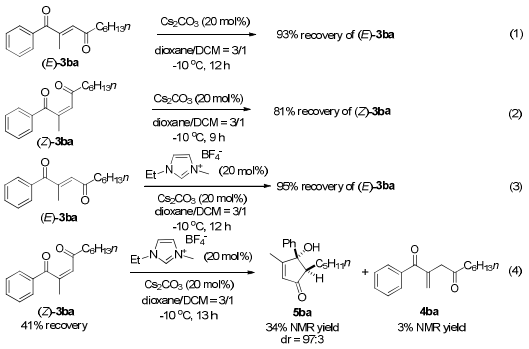
[a] Reaction conditions: A solution of NHC peecursor and Cs_2CO_3 in mixed solvent (5 mL) was stirred at rt for 10 min, then **1**, 1 mmol of **2**, and solvent (5 mL) were added sequentially. The resulting mixture was then stirred at T for the time indicated in the Table. [b] Determined by ^{1}H NMR. [c] 4*R**,5*R**-**5ca** (72%) and 4*R**,5*S**-**5ca** (3%) could be separated by column chromatography on silica gel. [d] 4*R**,5*R**-**5ee** (55%) and 4*R**,5*S**-**5ee** (1%) could be separated by column chromatography on silica gel. [e] The reaction was conducted at -10 °C for 32 h, then warmed up to 50°C for 14.5 h. 4% of (*E*)-**3bf**, 1% of **4bf**, and 4% of (*Z*)-**3bf** were also formed.

The reaction of aliphatic aldehyde **1f** was also studied. Nevertheless, the expected 4-alkyl substituted cyclopent-2-enone-4-ol was not formed, only 6% of **6fa** was observed, together with 58% recovery of aldehyde **1f** and 39% recovery of 1,2-allenone **2a** (Scheme 2). Obviously, the reactivity of aliphatic aldehyde towards NHC precursor **A1** is rather low in such a reaction.



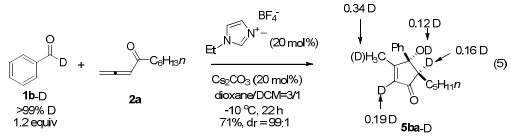
Scheme 2 The reaction of aliphatic aldehyde **1f**.

In order to gain insight into the reaction, possible intermediate (*E*)-**3ba** and (*Z*)-**3ba** were isolated utilizing the condition of entry 1 in Table S1 (see ESI†). As a result, cyclic product **5ba** couldn't be formed when starting from (*E*)-**3ba** under sole 20 mol % of Cs_2CO_3 (Scheme 3, eq 1) or under 20 mol % of **A1** and Cs_2CO_3, respectively (Scheme 3, eq 3). On the other hand, (*Z*)-**3ba** also couldn't be transferred to **5ba** catalyzed only by 20 mol % of Cs_2CO_3 (Scheme 3, eq 2). However, **5ba** was obtained in only 34% NMR yield (dr = 97:3) together with 3% of **4ba** when 20 mol % of **A1** and 20 mol % of Cs_2CO_3 were used as catalysts starting from (*Z*)-**3ba**, however, 41% recovery of (*Z*)-**3ba** was observed (Scheme 3, eq 4). These experiments indicated that there exist other pathways.



Scheme 3 Mechanistic studies.

An isotopic labeling experiments using deuterated benzaldehyde **1b**-D was conducted. As a result, deuterated product **5ba**-D was obtained, with 19% D-incorporation in the 2-position, 34% D-incorporation in the 3-position methyl group, 12% D-incorporation in the 4-position hydroxyl group, and 16% D-incorporation in the 5-position (eq 5).

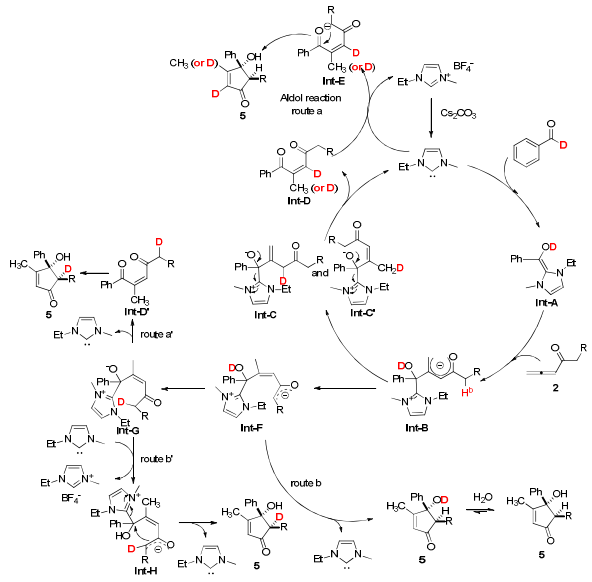


Based on the observation above, a possible mechanism was then proposed. *N*-heterocyclic carbene generated in situ reacted with aldehyde to give the Breslow intermediate **Int-A**, which would attack the central carbon of the allenone **2**, forming a delocalized anion intermediate **Int-B**.[18] After the formation of intermediate **Int-B**, four pathways are possible: (1) An intramolecular D-transfer occurred in intermediate **Int-B** led to intermediate **Int-C** and **Int-C'**. Then elimination of the NHC produced the deuterated 1,4-enediketones **Int-D**. With NHC acting as a Brønsted base,[18,19] the sequential Aldol reaction proceeded to give the final product with deuteration within the 2-position and the methyl group of 3-position (Scheme 4, route a); (2) H^b in **Int-B** may also transfer to the delocalized anion, giving enolate-type intermediate **Int-F**. Cyclic nucleophilic substitution occurred[20] to give the cyclic product **5** with the deuteration in the hydroxyl group, finally yielding the non-deuterated **5**, accompanied with the regeneration of NHC (Scheme 4, route b); (3) The deuterium in the hydroxyl group of **Int-F** may also transfer to trap the enolate, forming α-deuterated intermediate **Int-G**. Then a similar Aldol reaction like route a happened to give product **5** with deuteration at the 5-position via **Int-D'** (Scheme 4, route a'); (4) **Int-G** could also form intermediate **Int-H** followed by a pathway like route b to generate product **5** with deuteration at the 5-position (Scheme 4, route b'). Meanwhile, the *trans*-diastereoselectivity of products **5** may be explained by the stereo-electronic effect of oxygen atom in hydroxyl group adjacent to the NHC acting as a leaving group.

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Scheme 4 Possible reaction pathways.

In conclusion, we have developed an efficient diastereoselective synthetic route to useful cyclopent-2-enone-4-ols from readily available aromatic aldehydes and 1,2-allenones **2**. Mechanistic studies showed the essential role of NHC in this transformation. Further investigation is being carried out in our laboratory.

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